

NORVENICH, Germany

WMO: 105020

Lat: 50.8294N

Lon: 6.6603E

Elev: 134

StdP: 99.73

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.1	-10.6	1.5	-5.3	-8.4	1.9	-3.8	12.8	8.7	11.4	7.6	2.4	120	0.506

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.1	19.7	28.8	19.1	26.8	18.2	20.8	28.3	19.9	26.7	19.0	25.1	3.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.4	13.5	23.3	17.6	12.8	22.3	16.8	12.2	21.7	60.6	28.3	57.5	26.8	54.6	25.1	24.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.1	6.9	-10.8	35.2	3.8	1.7	-13.6	36.4	-15.8	37.4	-17.9	38.3	-20.7	39.5		
(5)			-11.1	22.8	3.7	1.0	-13.8	23.5	-16.0	24.1	-18.1	24.7	-20.8	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.0	3.1	4.0	6.8	10.2	14.1	17.1	19.3	18.8	15.3	11.4	7.0	3.9		
	DBStd	6.79	4.47	4.18	3.56	3.74	3.61	3.28	3.29	3.06	3.02	3.59	3.65	4.31		
	HDD10.0	867	216	171	108	43	6	0	0	0	1	27	102	193		
	HDD18.3	2834	472	402	356	243	137	62	28	31	99	216	339	448		
	CDD10.0	1218	2	2	11	50	134	214	287	272	161	69	12	3		
	CDD18.3	145	0	0	0	1	7	26	57	44	10	1	0	0		
	CDH23.3	1439	0	0	0	18	87	257	550	434	89	4	0	0		
	CDH26.7	458	0	0	0	1	16	72	192	157	20	0	0	0		
(14) Wind	WSAvg	3.0	3.6	3.5	3.4	2.9	2.7	2.7	2.8	2.6	2.7	2.8	3.0	3.3		
(15) Precipitation	PrecAvg	704	52	52	47	44	57	70	72	74	54	55	58	61		
	PrecMax	849	119	121	87	84	104	129	190	166	133	142	99	142		
	PrecMin	565	7	8	6	0	22	14	33	27	11	28	1	16		
	PrecStd	83	26	26	23	21	23	28	38	38	33	25	23	27		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.0	15.7	20.0	25.9	28.9	32.2	34.1	34.1	29.6	23.8	17.5	14.2		
		MCWB	10.9	9.9	11.9	14.6	18.4	21.0	20.1	20.5	19.0	17.0	13.6	11.1		
	2%	DB	12.1	12.9	17.0	22.7	25.6	28.8	31.1	30.9	25.9	20.9	15.1	12.9		
		MCWB	9.5	9.0	10.5	13.5	16.5	19.3	19.7	19.6	17.7	15.6	12.0	10.3		
	5%	DB	10.6	11.3	14.6	19.9	23.6	26.2	28.7	28.0	23.2	18.5	13.8	11.5		
		MCWB	8.3	8.3	9.9	12.7	15.8	18.0	19.0	18.8	16.7	14.5	11.1	9.1		
	10%	DB	9.0	10.0	12.7	17.2	21.2	24.0	26.4	25.3	20.9	16.8	12.1	10.0		
		MCWB	6.9	7.5	8.9	11.5	14.9	17.2	18.2	17.9	15.7	13.5	9.8	7.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.2	10.7	12.7	16.0	19.7	22.1	22.3	21.7	19.8	17.4	14.0	11.8		
		MCDB	13.5	13.7	17.8	22.8	26.3	29.6	30.3	30.3	26.4	21.8	16.7	13.6		
	2%	WB	9.7	9.6	11.4	14.4	18.0	20.3	20.9	20.5	18.5	16.0	12.3	10.5		
		MCDB	11.9	12.2	15.4	20.7	24.0	26.9	28.1	28.2	24.1	19.9	14.7	12.7		
	5%	WB	8.5	8.6	10.3	13.1	16.6	19.1	19.9	19.5	17.3	14.9	11.2	9.3		
		MCDB	10.5	11.1	13.8	18.5	21.7	24.9	26.8	26.3	22.1	18.0	13.5	11.2		
	10%	WB	7.0	7.5	9.3	12.0	15.3	17.8	19.0	18.5	16.2	13.9	10.1	8.0		
		MCDB	8.8	9.8	12.5	16.8	20.3	22.6	24.9	24.3	20.2	16.5	12.0	9.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.3	6.2	8.1	10.1	10.1	10.3	10.3	10.4	9.3	7.8	5.9	5.1		
		MCDBR	6.6	8.2	12.3	14.8	14.2	14.6	14.7	15.5	13.2	10.4	7.1	6.1		
	5% WB	MCWBR	5.2	5.6	7.2	7.4	7.0	7.0	5.9	6.4	6.5	6.2	5.1	4.7		
		MCWBR	6.4	7.4	10.1	12.8	12.3	12.5	12.8	12.9	11.5	9.7	6.8	5.6		
(40) Clear-Sky Solar Irradiance	taub		0.315	0.339	0.370	0.407	0.412	0.415	0.419	0.412	0.382	0.357	0.333	0.312		
		taud	2.432	2.385	2.312	2.238	2.258	2.268	2.255	2.302	2.367	2.436	2.431	2.409		
	Ebn at Noon		714	788	824	832	844	843	832	819	804	757	681	654		
		Edh at Noon	61	81	104	126	129	129	128	117	98	77	61	54		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.43	2.56	4.09	4.86	5.31	5.03	4.31	3.18	1.86	0.87	0.57		
	RadStd		0.12	0.27	0.39	0.47	0.45	0.49	0.61	0.42	0.39	0.18	0.12	0.08		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Nomenclature: See separate page